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Q1. String Character Frequency Counter

CODE

```
str=input("enter the charcter\n")
count={}
for ch in str:
    if ch!=" ":
        if ch in count:
            count[ch]+=1
        else:
            count[ch]=1
for ch in sorted(count):
    print(ch,'->',count[ch])
```

INPUT

hello world

OUTPUT

```
enter the charcter
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s=input("enter a sentence : ").lower()
v="aeiou"
vc=0
cc=0
vf={}
cf={}
for ch in s:
    if(ch.isalpha()):
        if(ch in v):
            vc+=1
            if ch in vf:
                vf[ch]+=1
            else:
                vf[ch]=1
        else:
            cc+=1
            if(ch in cf):
                cf[ch]+=1
            else:
                cf[ch]=1
print("vowels = ",v)
print("consonants = ",c)
if vf:
    maxv=max(vf,key=vf.get)
```

OUTPUT

(no output)

Q3. String Compression and Run-Length Encoding**CODE**

```
s=input()
compressed=""
i=0
while i<len(s):
    c=1
    while i+1<len(s) and s[i]==s[i+1]:
        i=i+1
        c=c+1
    if c==1:
        compressed=compressed+s[i]
    else:
        compressed=compressed +s[i]+str(c)
    i=i+1
if len(compressed)<len(s):
    print("Compressed:"+compressed)
    decompressed=""
    j=0
    while j<len(compressed):
        ch=compressed[j]
        j=j+1
        num=""
        while j<len(compressed) and compressed[j].isdigit():
            num=num+compressed[j]
            j=j+1
        if num=="":
            decompressed=decompressed+ch
        else:
            decompressed=decompressed+ch*int(num)
    print("Decompressed:"+decompressed)
else:
    print("Compressed not beneficial.")
    print("Original:"+s)
```

INPUT

aabbccdddee

OUTPUT

Compressed:a2b3c2d4e2
Decompressed:aabbccdddee

Q4. Advanced Text Statistics Analyser**CODE**

```
print()
```

OUTPUT